Project N.O.M.A.D. on Unraid

Public Version: 1.3

Updated: 09/04/2026

Document Information

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Assistance: **AI-supported drafting**

Project: Independent, community-created guide

Platform: Unraid + Docker Compose

Disclaimer

This manual documents the steps I personally used to deploy Project N.O.M.A.D. on Unraid. I am still learning Docker and Unraid, and this guide is provided as-is, with no guarantees.

I am not responsible for any issues, data loss, or misconfigurations that may occur.

I cannot provide active support beyond sharing what worked for me.

This guide is not affiliated with or endorsed by the Project N.O.M.A.D. team.

It was created independently, with the help of AI, to assist others who may be learning as well.

Change Log

Version 1.3 — 09/04/2026

- Added full **Change Log** section
- Updated version number to **1.3**

- Added CPU-only Ollama mode instructions
- Added instructions for removing GPU requirement while keeping Ollama
- **Document layout updated for clarity and consistency**
- Improved formatting and section structure
- Removed the **Quick Setup Guide** section (this is its own Document now as this was getting crowded)

Version 1.2 — 09/04/2026

- Added **external network creation step**
- Added expanded **Ollama removal / GPU disable** section
- Updated Troubleshooting for network issues
- General formatting improvements

Version 1.1 — 24/03/2026

- Added GPU acceleration instructions
- Added optional Ollama removal section
- Added folder structure improvements
- Added troubleshooting for MySQL and Redis
- Minor formatting fixes

Version 1.0 — Initial Release

- First public version of the manual
- Basic NOMAD deployment steps
- Initial compose file instructions
- No AI or GPU sections included

Table of Contents

1. Introduction
2. System Requirements
3. Folder Structure Preparation
4. Creating the External Docker Network
5. Preparing and Editing the Provided Compose File
6. Updating Required Fields and Uploading the File
7. Validating the Compose File (Optional)
8. Adding the Stack in Docker Compose Manager
9. Deploying the Stack

10. Verifying Container Status
11. Accessing the NOMAD Interface
12. Quick Guide: Enabling Ollama GPU Acceleration
13. How to Remove Ollama (AI) or Disable GPU Acceleration
14. Backup & Restore Strategy
15. Troubleshooting

1. Introduction

This manual provides a structured, step-by-step process for deploying Project N.O.M.A.D. on Unraid using Docker Compose.

It is designed for beginners and includes context for each step to support learning and troubleshooting.

Optional GPU-accelerated AI support is included via Ollama.

2. System Requirements

- Unraid server (6.12 or later recommended)
- Docker Compose Manager plugin installed
- Internet access
- Basic familiarity with Unraid
- Optional: NVIDIA GPU + NVIDIA Driver plugin (for AI acceleration)

3. Folder Structure Preparation

Purpose

Persistent data must be stored outside containers to survive updates and restarts.

Procedure

```
mkdir -p /mnt/user/appdata/nomad/storage
mkdir -p /mnt/user/appdata/nomad/mysql
mkdir -p /mnt/user/appdata/nomad/redis
mkdir -p /mnt/user/appdata/nomad/ollama
```

Result

Your folder structure is ready for NOMAD and Ollama.

4. Creating the External Docker Network

(Added in Version 1.2)

Purpose

The compose file declares the network as **external**, so Docker will not create it automatically.

This network must exist before the stack is deployed.

Procedure

Run:

```
docker network create project-nomad_default
```

If the network already exists, Docker will say so.

Result

A persistent external network named `project-nomad_default` is created.

5. Preparing and Editing the Provided Compose File

Purpose

You will edit the compose file locally, then upload it to Unraid.

Procedure

1. Open the compose file in a YAML-friendly editor.
2. Ensure the network block is:

```
networks:
  default:
```

```
    name: project-nomad_default
    external: true
```

3. Review all services.
4. Save the file as:

```
docker-compose.yml
```

6. Updating Required Fields and Uploading the File

Fields to Update

- `MYSQL_ROOT_PASSWORD`
- `MYSQL_PASSWORD`
- `APP_KEY`

Upload Path

Upload the file to:

```
/mnt/user/appdata/nomad/
```

7. Validating the Compose File (Optional)

```
cd /mnt/user/appdata/nomad
docker compose config
```

8. Adding the Stack in Docker Compose Manager

Stack Name:

```
project-nomad
```

Compose File Path:

```
/mnt/user/appdata/nomad/docker-compose.yml
```

9. Deploying the Stack

Click **Up** to deploy.

10. Verifying Container Status

Expected containers:

- nomad_admin
- nomad_mysql
- nomad_redis
- nomad_updater
- nomad_dozzle
- nomad_ollama (AI service)

11. Accessing the NOMAD Interface

http://YOUR_UNRAID_IP:8081

12. Quick Guide: Enabling Ollama GPU Acceleration

Step 1 — Install NVIDIA Driver Plugin

Step 2 — Confirm Ollama Service Exists

Step 3 — Pull a Model

```
docker exec -it nomad_ollama ollama pull llama2
```

Step 4 — Restart Stack

Step 5 — Verify GPU Acceleration

13. How to Remove Ollama (AI) or Disable GPU Acceleration

(Expanded in Version 1.3)

You now have **three options**:

A. Remove Ollama Completely

Delete the entire `ollama:` block from your compose file.

(Optional) Remove data:

```
rm -rf /mnt/user/appdata/nomad/ollama
```

Redeploy the stack.

B. Keep Ollama but Disable GPU (CPU-Only Mode)

(New in Version 1.3)

If you want Ollama to run **without requiring a GPU**, remove the GPU reservation.

Step 1 — Remove this block:

```
deploy:
  resources:
    reservations:
      devices:
        - capabilities: [gpu]
```

Step 2 — Redeploy the stack

Down → Up

Step 3 — Confirm CPU mode

```
docker logs nomad_ollama
```

You should **not** see CUDA/GPU initialization.

Result

- Ollama works normally
- NOMAD can still use AI
- No GPU required

C. Re-Enable GPU Later

Re-add:

```
deploy:
  resources:
    reservations:
      devices:
        - capabilities: [gpu]
```

Redeploy the stack.

14. Backup & Restore Strategy

Back up:

```
/mnt/user/appdata/nomad/
```

Use **CA Backup/Restore Appdata**.

15. Troubleshooting

Includes fixes for:

- Missing network
- YAML errors
- MySQL failures
- Permission issues
- Port conflicts
- AI/GPU issues

End of Manual

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